

Emiliano M. Silva

Researcher in Sustainable Extractive Metallurgy

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Professional Summary

Forensic Expert and Sustainable Mining Researcher with over 10 years of experience bridging field investigations with scientific analysis. Expert in applying forensic methodologies, from crime scene processing to environmental field data collection and analysis, to ensure technical integrity in both criminal justice and mining sectors. Proven track record in managing regional forensic units and leading data-driven research to translate complex field data into actionable insights for regulatory bodies, legal proceedings, and institutional governance.

Chemist, Civil Engineer, and Ph.D. candidate in Mining Engineering with interdisciplinary expertise in hydrometallurgy, mineral processing, analytical chemistry, environmental geochemistry, and sustainable extractive metallurgy. My research focuses on mercury-free gold extraction, alternative leaching technologies, metallurgical process optimization, and mine waste valorization. I have conducted research and field campaigns in Brazil, Suriname, and Colombia while collaborating internationally with the University of British Columbia. My twelve years as an official Crime Scene Investigator have strengthened my skills in field investigations, systematic data collection, quality assurance, and multidisciplinary problem solving.

Research Interests

Sustainable Hydrometallurgy • Gold Leaching Technologies • Critical Metals Recovery • Process Mineralogy • Extractive Metallurgy • Circular Economy • Mine Waste Valorization • Process Engineering

Education

2023–Present **Ph.D. in Mineral Engineering**, *University of São Paulo (USP)*, São Paulo, Brazil

Developing sustainable processing pathways for Artisanal Gold Mining (AGM) by replacing mercury with cleaner alternative technologies. The research focuses on process optimization to simultaneously increase gold recovery rates and reduce operational costs through resource-efficient mineral processing techniques.

2020–2022 **B.Sc. in Civil Engineering**, *Faculdade Anhanguera*, Mato Grosso, Brazil

Focused on construction management and resource economy in housing projects.

2017–2018 **Specialization in Criminal Forensics**, *Faculdade Poliensino*, Mato Grosso, Brazil

Postgraduate studies in criminalistics and forensic methodologies, specialized on crime scene investigation protocols.

- 2010–2012 **M.Sc. in Geosciences**, *Federal University of Mato Grosso (UFMT)*, Mato Grosso, Brazil
Dissertation utilized exploratory data analysis to evaluate surface water quality as a geochemical indicator of selenium leaching in mining environments. Research included an international fellowship at the University of British Columbia (UBC), Canada, funded by the Emerging Leaders in the Americas Program (ELAP). Focused on the statistical modeling of environmental parameters and geochemical behavior in coal exploration zones.
- 2005–2009 **B.Sc. in Chemistry**, *Federal University of Mato Grosso (UFMT)*, Mato Grosso, Brazil
Conducted undergraduate research focused on analytical chemistry and the continuous monitoring of mercury levels in regional ecosystems. Gained two years of technical experience at the University's Fuel Analytical Center, performing quality control and quantitative analysis of fuels to ensure regulatory compliance and technical conformity.

Professional Experience

- 2023–Present **Researcher**, *NAP.Mineração – Research Center for Responsible Mining*, University of São Paulo (USP), Brazil
Collaborate on cross-border research initiatives focused on sustainable artisanal and small-scale gold mining (ASGM) across Brazil, Colombia and Suriname. Execute on-site diagnostics, environmental sampling, and data analysis to optimize gold recovery and resource efficiency. Contribute to the identification of systemic gaps in extraction methodologies and the development of strategies to ensure compliance with international standards.
- 2022–Present **Construction Engineer**, *Independent Practice*, Mato Grosso, Brazil
Directed the field execution and financial management of 10+ residential construction projects to date. Specialized in resource economy and cost control, maximizing budgetary efficiency through the strategic allocation of materials and site supervision to minimize waste and reduce project costs.
- 2014–Present **Official Forensic Expert**, *POLITEC – State Secretariat of Public Security*, Mato Grosso, Brazil
Apply rigorous scientific methodologies to conduct official forensic investigations, with extensive field processing of complex crime scenes including homicides, traffic accidents, environmental offenses, forensic ballistics, and narcotics. Generate defensible, objective technical reports for legal proceedings through detailed evidence analysis. Served as Regional Manager for units in Juína and Primavera do Leste, Mato Grosso, overseeing operations, ensuring strict chain-of-custody compliance, and acting as the primary liaison with judicial and regulatory authorities.
- 2017–2020 **Part-time Lecturer and Coordinator**, *Faculdade Poliensino*, Mato Grosso, Brazil
Coordinated and delivered specialized curricula for the Post-Graduate program (Specialization) in Criminalistics and Forensic Sciences. Focused on providing advanced technical training dedicated to specializing professionals in crime scene processing and forensic investigative methodologies.
- 2012–2014 **Full-time Lecturer**, *Centro Universitário de Várzea Grande (UNIVAG)*, Mato Grosso, Brazil
Instructed core STEM disciplines including Statistics, Materials Science, General and Analytical Chemistry. Served as a member of the institutional accreditation team for the Civil Engineering program, managing the formal certification process by the Brazilian Ministry of Education (MEC).
- 2011–2014 **Guest Lecturer**, *Universidade do Estado de Mato Grosso (UNEMAT)*, Mato Grosso, Brazil
Delivered in-class and laboratory-based modules in General and Analytical Chemistry. Focused on chemical analysis methodologies and scientific standards for chemistry degree candidates.
- 2011–2011 **Junior Geochemist (Trainee)**, *Lorax Environmental Services Ltd.*, Vancouver, Canada
Performed as an Emerging Leaders in the Americas Program (ELAP) scholar. Evaluated surface water quality and geochemical data from multiple mining operations, utilizing statistical analysis to process complex datasets for environmental monitoring and regulatory compliance.

2010–2011 **Assistant Professor**, *Federal University of Mato Grosso (UFMT)*, Mato Grosso, Brazil
 Taught in-class and laboratory-based disciplines in Applied Analytical Chemistry and General Chemistry for Civil, Sanitary, and Environmental Engineering programs. Managed experimental sessions and mentored undergraduate students in chemical analysis and laboratory techniques.

Core Competencies

	Level	Skill	Years	Comment
Technical	■■■■■	Field Operations	15+	<i>Expert Forensic and Environmental Sampling</i>
	■■■■■	Analytical Chemistry	15+	<i>Environmental Monitoring of Trace Elements</i>
	■■■■■	Technical Reporting	12+	<i>Scientific and Judicial Documentation</i>
	■■■■■	Data Analysis	12+	<i>Statistical Modeling: Excel, R, Python</i>
	■■■■■	AI-Assisted Development	2+	<i>Rapid Prototyping and Data Automation</i>
Leadership	■■■■■	Regional Unit Management	10+	<i>Forensic Operations and Unit Command</i>
	■■■■■	Construction Management	3+	<i>Resource Optimization and Work Oversight</i>
Languages	■■■■■	Portuguese	Native	
	■■■■■	English	15+	<i>Duolingo English Test Score 130 (2026/07/03)</i>
	■■■■■	Spanish	10+	<i>Field Work Proficiency</i>
	■■■■■	Italian	< 1	<i>Currently studying</i>

Selected Publications

- [1] Anthony Druiventak et al. "From Byproduct to Business: A Conceptual Micro-Enterprise Model for Artisanal and Small-Scale Gold Mining (ASGM) Tailings in Suriname". In: *The Extractive Industries and Society* 27 (2026), p. 101931. DOI: 10.1016/j.exis.2026.101931. URL: <https://doi.org/10.1016/j.exis.2026.101931>.
- [2] Emiliano M. Silva, Marcello M. da Veiga, and Giorgio De Tomi. "System-Wide Evaluation of Gold and Silver Recovery and Mercury Losses in an Artisanal Amalgamation–Cyanidation Circuit: A Diagnostic Approach to Mercury-Free Processing". In: *Minerals* 16.8 (2026), p. 784. DOI: 10.3390/min16080784. URL: <https://doi.org/10.3390/min16080784>.
- [3] Emiliano M. Silva et al. "From Agro-Industrial Waste to Gold Lixiviant: Evaluating Cassava Wastewater Applications in Artisanal Mining". In: *Mining* 5 (2025), p. 64. DOI: 10.3390/mining5040064. URL: <https://doi.org/10.3390/mining5040064>.
- [4] Emiliano M. Silva et al. "Production of cement-tailings bricks with artisanal gold mining waste". In: *Cleaner Waste Systems* 11 (2025), p. 100269. DOI: 10.1016/j.clwas.2025.100269. URL: <https://doi.org/10.1016/j.clwas.2025.100269>.
- [5] Marcello M. Veiga, N. C. Anene, and Emiliano M. Silva. "Four Decades of Efforts to Reduce or Eliminate Mercury Pollution in Artisanal Gold Mining". In: *Minerals* 15.4 (2025), p. 376. DOI: 10.3390/min15040376. URL: <https://doi.org/10.3390/min15040376>.



Updated information available at:
<https://emilianomsilva.github.io/>